

Implementation Phase Plan

Device Monitoring Application — Android, Multi-Tenant

Phase 1: Foundation Setup

Folder structure (android-app, backend, dashboard, docs); Backend project init (Node.js + Express skeleton); PostgreSQL setup + connection config; Git repo structure; environment config templates.

Phase 2: Database Schema & Backend Core

Tables: tenants, users (admin), devices, feature_flags, location_logs, app_usage_logs, sms_logs, call_logs, installed_apps, alerts, geofences. Multi-tenant row-level isolation. JWT auth. Core CRUD APIs: tenant management, device registration, user management.

Phase 3: Android App — Core Skeleton

Kotlin project setup, MVVM architecture. Consent screen UI (mandatory first-launch). Permission request flow. Device registration API integration. Room DB setup (local buffer).

Phase 4: Data Collection Modules (Android)

App usage tracking (UsageStatsManager + WorkManager). SMS log module. Call log module (metadata only). Installed apps list. Website history module. Each: collect -> Room buffer -> sync API.

Phase 5: Location & Geofencing

On-demand location fetch (FCM-triggered). Native GeofencingClient integration. Geofence CRUD from dashboard -> sync to device. Alert trigger flow (boundary cross -> backend -> dashboard).

Phase 6: Remote Actions

FCM command dispatch system (backend). Remote Lock (Device Admin API). File Manager (Storage Access Framework listing). File Download (on-demand WebSocket streaming, no server storage).

Phase 7: App/Website Blocking

Accessibility Service / Device Admin blocking logic. Blocklist/allowlist + schedule sync from dashboard.

Phase 8: Admin Dashboard (React)

Login/auth UI. Tenant & device management pages. Reports: location map, usage charts, SMS/call log tables, installed apps. Feature flag toggle UI. Remote action UI (lock, file manager, download). Alerts panel.

Phase 9: Integration & Testing

End-to-end testing: install -> consent -> data flow -> dashboard display. Battery/background execution testing on real device. Multi-tenant isolation testing (data leak check).

Phase 10: Deployment

Backend deploy (AWS/DigitalOcean). APK build & signing. Dashboard deploy. Production DB migration.

Execution Order

Phase 1 -> Phase 2 (backend + DB foundation first, since the Android app needs APIs to sync against) -> Phases 3-7 (Android modules) -> Phase 8 (Dashboard) -> Phases 9-10 (Testing & Deployment).